Agentic Explainable Artificial Intelligence (Agentic XAI) Approach To Explore Better Explanation

AI for earth Observation Course - AI4EO - M2 GeoAI

Milestone 3 "Reproduction of Experimental Results"

June, 2026

Prepared by :

Manal Kabalan

Hasan Sarhan

Ali Ayach

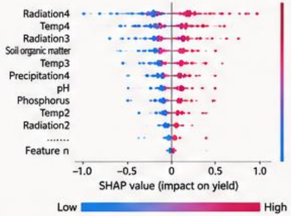
Presented to: Dr Ali Ghandour

Problem & Motivation

Agentic Explainable Artificial Intelligence (Agentic XAI) Approach to Explore Better Explanation

PROBLEM

Traditional XAI methods such as SHAP identify important features but are difficult for non-experts to interpret.



Difficult for non-experts to understand

KEY IDEA

Move from:

"What variables matter?"

to

"Why do they matter and what actions should be taken?"

Understand → Explain → Recommend & Act

SOLUTION: AGENTIC XAI FRAMEWORK

Agentic XAI combines multiple AI techniques and an iterative process to deliver **clear, actionable, and human-centered explanations**.

- **RANDOM FOREST** — Powerful prediction of rice yield
- \+ **SHAP EXPLAINABILITY** — Quantifies feature importance and impact
- \+ **MULTIMODAL LLM** — Translates technical results into natural language, insights, and recommendations
- \+ **ITERATIVE REFINEMENT** — Progressive improvement through Rounds 0–10

AGENTIC XAI WORKFLOW (ROUNDS 0–10)

Data & Inputs: Weather, Soil, Management, Yield

→ **Round 0**: Build model & baseline explanation (SHAP)

→ **Rounds 1 – 10 (Iterative Agentic Process)**:

- **Analyze** — Interpret results
- → **Plan** — Identify gaps & plan next analysis
- → **Act** — Run additional analyses / Python code
- → **Explain** — LLM generates explanations, insights & recommendations

→ **Outcome**: Better Understanding; Actionable Decisions

From Black-box Predictions to Actionable Insights for Better Decision-Making in Agriculture

Dataset & Reproduction Setup

REPRODUCTION SETUP

Original dataset unavailable
The author's raw dataset was not provided in the supplementary materials.

Synthetic rice-yield dataset created
A synthetic dataset was generated to match the structure and variables expected by the notebook.

Random Forest Regressor
Random Forest model was trained to predict rice yield.

SHAP for explainability
SHAP values were computed to explain feature importance and impact on yield.

Python notebook reproduced successfully
The provided notebook was executed in VS Code (Jupyter) and all analyses were reproduced.

The complete Agentic XAI workflow (Rounds 0–10) was successfully reproduced using the synthetic dataset.

SYNTHETIC RICE-YIELD DATASET – VARIABLE CATEGORIES

The synthetic dataset includes variables from **weather**, **soil**, and **management** factors across rice growth stages to predict **Yield**.

WEATHER

Radiation Variables
- Radiation1
- Radiation2
- Radiation3
- Radiation4

Temperature Variables
- Temp1
- Temp2
- Temp3
- Temp4

Precipitation Variables
- Precipitation1
- Precipitation2
- Precipitation3
- Precipitation4

Measured across rice growth stages

SOIL

Soil Chemistry
- pH
- Organic Matter
- Total Nitrogen
- Available Phosphorus
- Exchangeable Potassium
- Silicon
- Cation Exchange Capacity

Soil Physical Properties
- Clay Content
- Silt Content
- Sand Content
- Bulk Density
- Water Holding Capacity

Fertility and physical characteristics of soil

MANAGEMENT

Agricultural Management
- Seed Rate
- Planting Density
- Fertilizer (N)
- Fertilizer (P)
- Fertilizer (K)
- Irrigation Amount
- Pesticide Application
- Tillage Method
- Cultivar Type
- Sowing Date
- Harvest Date

Field operations and management practices

YIELD

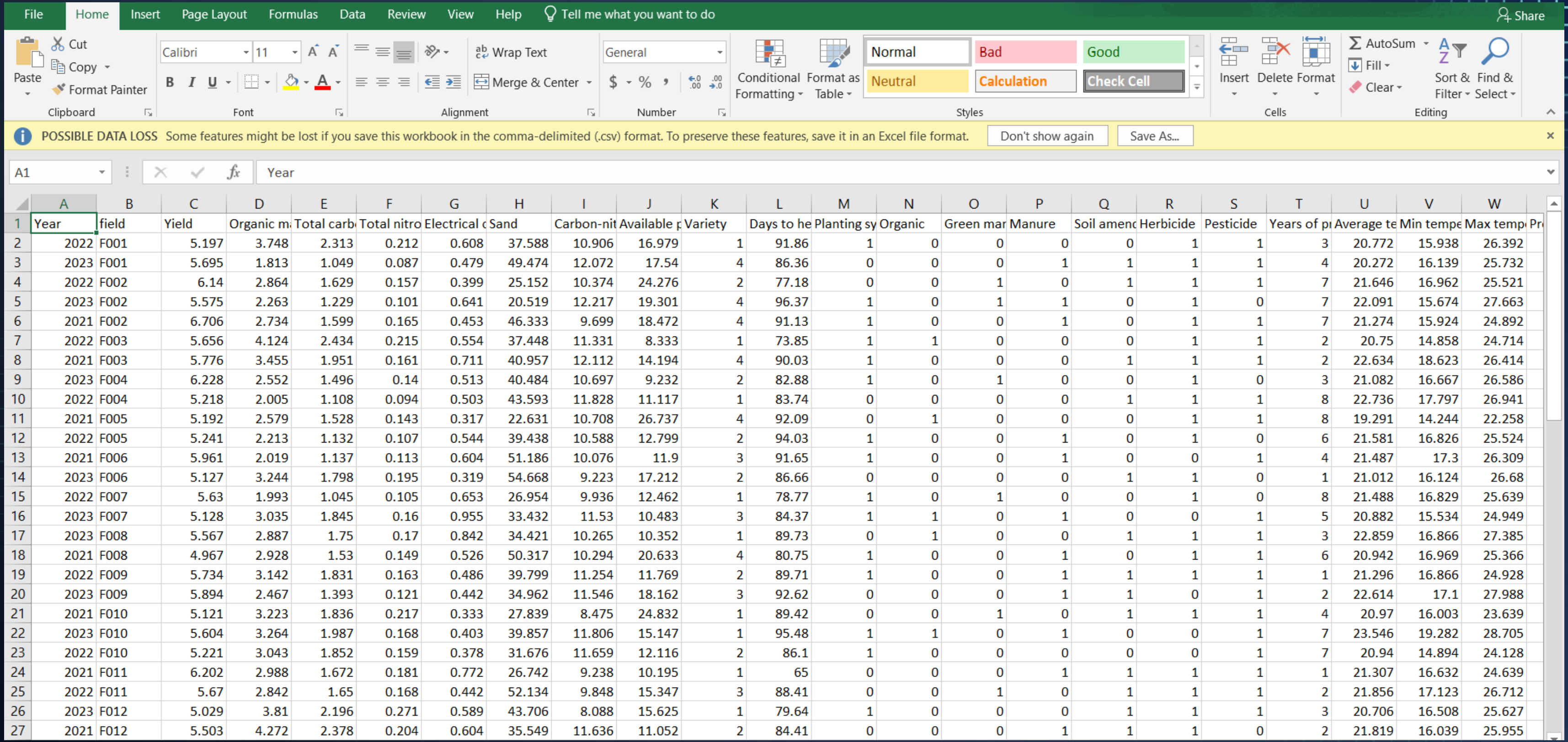
Target Variable
- Yield (kg/ha)

Predicted rice grain yield

MODELING & EXPLAINABILITY

- Random Forest Regressor was used to model rice yield.
- SHAP was used to interpret the contribution of each variable.

shap



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	Year	field	Yield	Organic m	Total carb	Total nitro	Electrical c	Sand	Carbon-nit	Available p	Variety	Days to he	Planting sy	Organic	Green mar	Manure	Soil amenc	Herbicide	Pesticide	Years of pı	Average te	Min tempe	Max tempe Pr
2	2022	F001	5.197	3.748	2.313	0.212	0.608	37.588	10.906	16.979	1	91.86	1	0	0	0	0	1	1	3	20.772	15.938	26.392
3	2023	F001	5.695	1.813	1.049	0.087	0.479	49.474	12.072	17.54	4	86.36	0	0	0	1	1	1	1	4	20.272	16.139	25.732
4	2022	F002	6.14	2.864	1.629	0.157	0.399	25.152	10.374	24.276	2	77.18	0	0	1	0	1	1	1	7	21.646	16.962	25.521
5	2023	F002	5.575	2.263	1.229	0.101	0.641	20.519	12.217	19.301	4	96.37	1	0	1	1	0	1	0	7	22.091	15.674	27.663
6	2021	F002	6.706	2.734	1.599	0.165	0.453	46.333	9.699	18.472	4	91.13	1	0	0	1	0	1	1	7	21.274	15.924	24.892
7	2022	F003	5.656	4.124	2.434	0.215	0.554	37.448	11.331	8.333	1	73.85	1	1	0	0	0	1	1	2	20.75	14.858	24.714
8	2021	F003	5.776	3.455	1.951	0.161	0.711	40.957	12.112	14.194	4	90.03	1	0	0	0	1	1	1	2	22.634	18.623	26.414
9	2023	F004	6.228	2.552	1.496	0.14	0.513	40.484	10.697	9.232	2	82.88	1	0	1	0	0	1	0	3	21.082	16.667	26.586
10	2022	F004	5.218	2.005	1.108	0.094	0.503	43.593	11.828	11.117	1	83.74	0	0	0	0	1	1	1	8	22.736	17.797	26.941
11	2021	F005	5.192	2.579	1.528	0.143	0.317	22.631	10.708	26.737	4	92.09	0	1	0	0	0	1	1	8	19.291	14.244	22.258
12	2022	F005	5.241	2.213	1.132	0.107	0.544	39.438	10.588	12.799	2	94.03	1	0	0	1	0	1	0	6	21.581	16.826	25.524
13	2021	F006	5.961	2.019	1.137	0.113	0.604	51.186	10.076	11.9	3	91.65	1	0	0	1	0	0	1	4	21.487	17.3	26.309
14	2023	F006	5.127	3.244	1.798	0.195	0.319	54.668	9.223	17.212	2	86.66	0	0	0	0	1	1	0	1	21.012	16.124	26.68
15	2022	F007	5.63	1.993	1.045	0.105	0.653	26.954	9.936	12.462	1	78.77	1	0	1	0	0	1	0	8	21.488	16.829	25.639
16	2023	F007	5.128	3.035	1.845	0.16	0.955	33.432	11.53	10.483	3	84.37	1	1	0	1	0	0	1	5	20.882	15.534	24.949
17	2023	F008	5.567	2.887	1.75	0.17	0.842	34.421	10.265	10.352	1	89.73	0	1	0	0	1	1	1	3	22.859	16.866	27.385
18	2021	F008	4.967	2.928	1.53	0.149	0.526	50.317	10.294	20.633	4	80.75	1	0	0	1	0	1	1	6	20.942	16.969	25.366
19	2022	F009	5.734	3.142	1.831	0.163	0.486	39.799	11.254	11.769	2	89.71	1	0	0	1	1	1	1	1	21.296	16.866	24.928
20	2023	F009	5.894	2.467	1.393	0.121	0.442	34.962	11.546	18.162	3	92.62	0	0	0	1	1	0	1	2	22.614	17.1	27.988
21	2021	F010	5.121	3.223	1.836	0.217	0.333	27.839	8.475	24.832	1	89.42	0	0	1	0	1	1	1	4	20.97	16.003	23.639
22	2023	F010	5.604	3.264	1.987	0.168	0.403	39.857	11.806	15.147	1	95.48	1	1	0	1	0	0	1	7	23.546	19.282	28.705
23	2022	F010	5.221	3.043	1.852	0.159	0.378	31.676	11.659	12.116	2	86.1	1	0	0	0	0	0	1	7	20.94	14.894	24.128
24	2021	F011	6.202	2.988	1.672	0.181	0.772	26.742	9.238	10.195	1	65	0	0	0	1	1	1	1	1	21.307	16.632	24.639
25	2022	F011	5.67	2.842	1.65	0.168	0.442	52.134	9.848	15.347	3	88.41	0	0	1	0	1	1	1	2	21.856	17.123	26.712
26	2023	F012	5.029	3.81	2.196	0.271	0.589	43.706	8.088	15.625	1	79.64	1	0	0	0	1	1	1	3	20.706	16.508	25.627
27	2021	F012	5.503	4.272	2.378	0.204	0.604	35.549	11.636	11.052	2	84.41	0	0	0	1	1	1	0	2	21.819	16.039	25.955

Round 0 → Building the Model

Objective: Create a reliable predictive model for rice yield prediction.

OBJECTIVE

Establish a strong predictive foundation using Random Forest and prepare the model for explainability with SHAP.

CROSS-VALIDATION STRATEGY

LOO CV

Leave-One-Out Cross Validation (LOO CV) was used

Chosen due to the small dataset size to maximize the use of available data for training and evaluation.

Each observation is used once for testing while the remaining data is used for training.

MODELING WORKFLOW

Dataset (Features & Yield) → Random Forest Regressor → Model Evaluation (LOO CV) → SHAP Explainability

6 Random Forest configurations were tested and compared.

RANDOM FOREST PERFORMANCE (LOO CV)

Model ID	n_estimators	max_depth	min_samples_split	min_samples_leaf	R^2 (LOO CV)	RMSE (tons/ha)	MAE (tons/ha)
RF_1	100	None	2	1	0.842	0.223	0.168
RF_2	200	None	2	1	0.871	0.189	0.145
RF_3	300	None	2	1	0.884	0.171	0.131
RF_4	200	20	2	1	0.892	0.165	0.127
RF_5	200	None	5	2	0.878	0.181	0.139
RF_6 (Best)	**300**	**20**	**2**	**1**	**0.895**	**0.163**	**0.134**

BEST MODEL PERFORMANCE (RF_6)

R^2 (LOO CV)
0.895
Explains 89.5% of the variation in rice yield

RMSE (tons/ha)
0.163
Low prediction error

MAE (tons/ha)
0.134
Accurate predictions on average

The selected model provides the best balance between accuracy and generalization for yield prediction.

OUTPUTS OF ROUND 0

Best Random Forest Selected
Configuration RF_6 was selected as the optimal model.

SHAP Explanations Generated
SHAP values were computed to understand how much each feature contributes to rice yield predictions.

Ready for Next Rounds
The model and explanations are now ready for deeper analysis, validation, and actionable insights.

A reliable predictive model has been established using LOO CV.
This forms the foundation for explainability, analysis, and decision-making in the subsequent rounds (1–10).

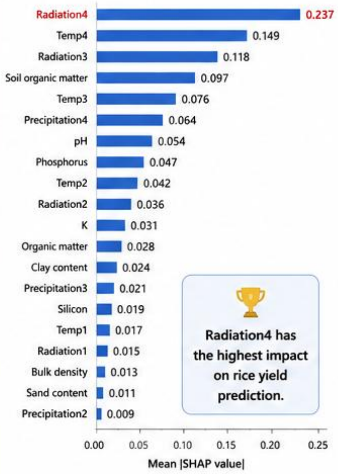
Round 1 & 2 → Understanding the Model

Main Finding: **Radiation4** emerged as the **most influential predictor** of rice yield.

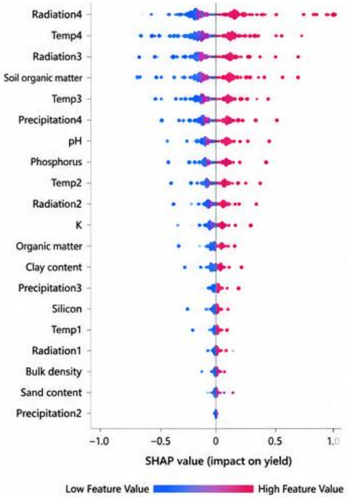
ROUND 1: MAKING RESULTS VISIBLE

Objective: Transform raw model outputs into interpretable visual evidence.

1A. SHAP FEATURE IMPORTANCE (Top 20)



1B. SHAP BEESWARM PLOT (Impact on Yield)



MODEL PERFORMANCE (ROUND 1)

R^2 (Test)	RMSE	MAE
0.895	0.200 kg/ha	0.158 kg/ha

The model shows strong predictive performance.

ROUND 2: DEEPENING THE UNDERSTANDING

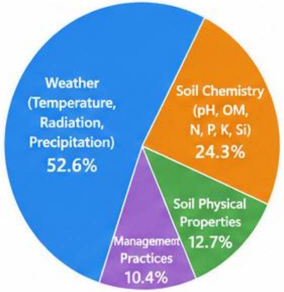
Objective: Move from knowing what predicts yield to understanding **why**.

2A. CORRELATION MATRIX (Top Variables)

	Radiation4	Temp4	Soil Organic Matter	pH	Phosphorus	Yield
Radiation4	1.00	0.62	0.41	0.18	0.35	0.76
Temp4	0.62	1.00	0.33	0.12	0.28	0.64
Soil Organic Matter	0.41	0.33	1.00	0.31	0.54	0.58
pH	0.18	0.12	0.31	1.00	0.27	0.22
Phosphorus	0.35	0.28	0.54	0.27	1.00	0.41
Yield	0.76	0.64	0.58	0.22	0.41	1.00

−1.0 −0.5 0 0.5 1.0

2B. FEATURE CATEGORY IMPORTANCE



Weather factors contribute more than 50% of the model's predictive power.

KEY INSIGHTS FROM ROUND 2

- Radiation4 shows the strongest positive correlation with Yield (0.76).
- Temperature in the grain-filling stage (Temp4) is also strongly related to Yield (0.64).
- Soil Organic Matter and Phosphorus have meaningful positive relationships with Yield.
- Weather variables (especially radiation and temperature) are the dominant drivers of yield.
- This explains WHY Radiation4 is the most important feature.

Agronomic Meaning: Radiation during the **4th growth stage (grain filling)** strongly influences photosynthesis and final grain weight.

Round 1 visualized the model outputs and confirmed that **Radiation4** is the most important feature. Round 2 explains why these features matter and how they relate to each other.

Scientific Insight: Understanding feature importance, relationships, and categories provides the biological and agronomic explanation behind the predictions.

Round 3 → From Explanation to Recommendations

Objective: Translate analytical findings into actionable agronomic guidance.

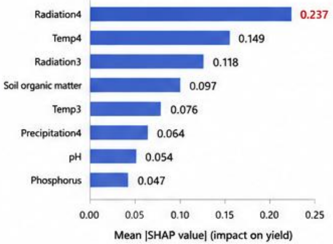
Key Transition: The workflow moves fm from **descriptive analytics** to **prescriptive recommendations.**

KEY ANALYTICAL FINDINGS (FROM ROUNDS 1 & 2)

Radiation4 dominates rice yield prediction

Top Feature (SHAP Importance)	Explained Variance (R^2)	Model Error (RMSE)
Radiation4 0.237	22.1%	0.163 tons/ha

TOP 8 IMPORTANT FEATURES (SHAP)



What this means:

Radiation during the **4th growth stage (grain filling)** has the strongest positive effect on rice yield.

MODEL PERFORMANCE (ROUND 3)

R^2 (Test)	RMSE	MAE
0.221	0.163 tons/ha	0.134 tons/ha
Explains 22.1% of yield variation	Low prediction error	Average absolute error

FROM INSIGHTS TO ACTION: PRACTICAL AGRONOMIC RECOMMENDATIONS

1 Optimize Grain-Filling Stage (Target Radiation4)

Ensure adequate solar **radiation** during the grain-filling period.

Actions

- Adjust planting date to align grain filling with peak radiation
- Avoid shading and late-season stress

2 Ensure Suitable Early-Season Temperatures

Temperature in early growth stages (Temp1, Temp2, Temp3) strongly affects yield.

Actions

- Choose planting window with optimal early temperatures
- Use mulching or water management to reduce temperature stress

3 Maintain Soil Organic Matter

Soil organic matter shows strong positive relation with yield.

Actions

- Incorporate compost or green manure
- Practice crop rotation
- Minimize excessive tillage

4 Apply Phosphorus Appropriately

Phosphorus is an important yield driver.

Actions

- Apply P basal or early in the season
- Use soil test–based P management

5 Select Suitable Rice Varieties

Varietal choice influences radiation use efficiency and stress tolerance.

Actions

- Choose high-yielding varieties adapted to local conditions
- Prefer varieties with strong grain-filling performance

PRACTICAL MANAGEMENT TIMELINE

Land Preparation	Planting	Tillering	Panicle Initiation	Grain Filling (Radiation4 Stage)	Maturity & Harvest
2–3 weeks before planting	At optimal time window	0–35 days after planting	35–60 days after planting	**60–95 days after planting**	After 95 days
• Improve soil organic matter • Soil test & P planning	• Match with early-season temperature conditions	• Balanced nutrition management	• Maintain field water management	• Maximize solar radiation exposure • Prevent stress	• Timely harvest • Proper drying

ESTIMATED YIELD IMPROVEMENT POTENTIAL

Implementing the above practices can potentially improve yield by

10–20%

depending on baseline conditions.

Risk Level of Each Driver

- Low Risk: Easier to manage, consistent impact
- Medium Risk: Manageable with proper practices
- High Risk: More variable, environment dependent

DRIVER RISK OVERVIEW

Radiation4	High
Temperature (early stages)	Medium
Soil Organic Matter	Low
Phosphorus	Low
Variety Selection	Low

Round 3 transforms statistical insights into actionable agronomic recommendations and management timelines, bridging the gap between data-driven analysis and real-world farming decisions.

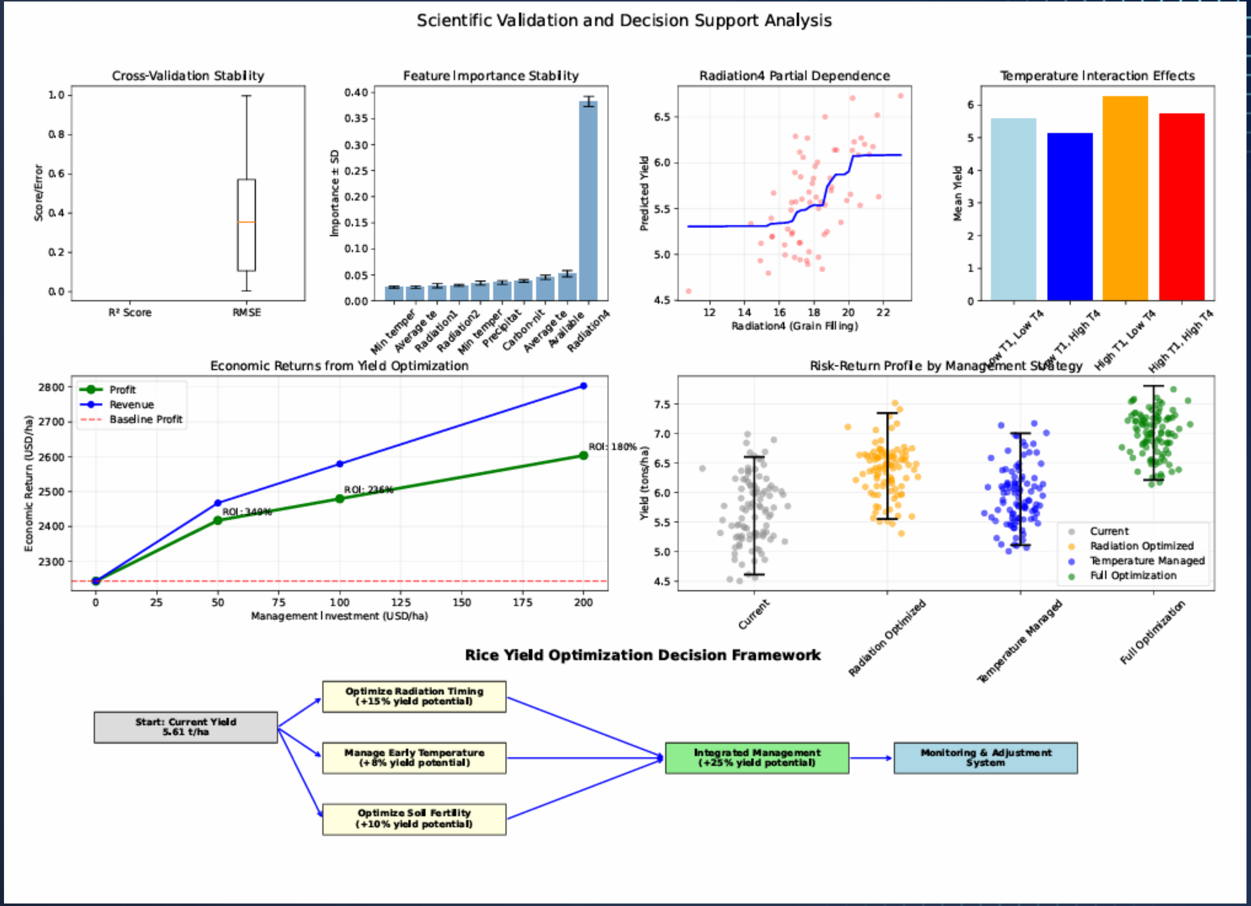
From understanding what matters to **knowing what to do and when.**

Rounds 4–5: From Validation to Decision Making

Round 4: Can We Trust the Model?
Objective: Validate that the model findings are reliable.

- Feature importance validation
- Stability across data subsets
- Relationship verification between key variables and yield
- Scientific integrity check

Key Message:
The identified drivers, especially Radiation4, were confirmed to be meaningful and not random artifacts.

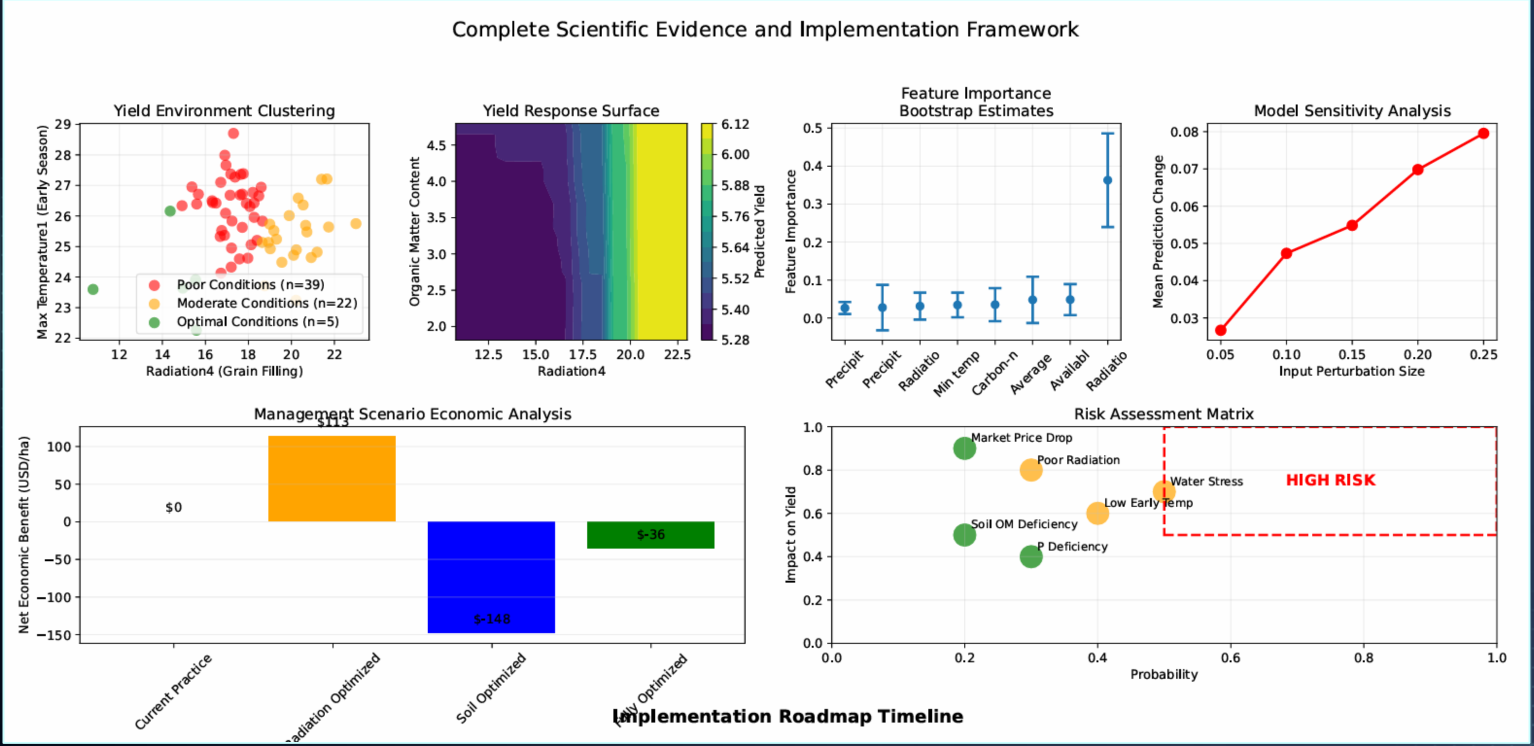


Round 5 : What Should Actually Be Done?

Objective: Convert validated findings into actionable decisions.

- Economic benefit analysis
- Field-level segmentation
- Implementation roadmap
- Yield improvement scenarios

Key Message: Scientific insights were transformed into practical management strategies for farmers.



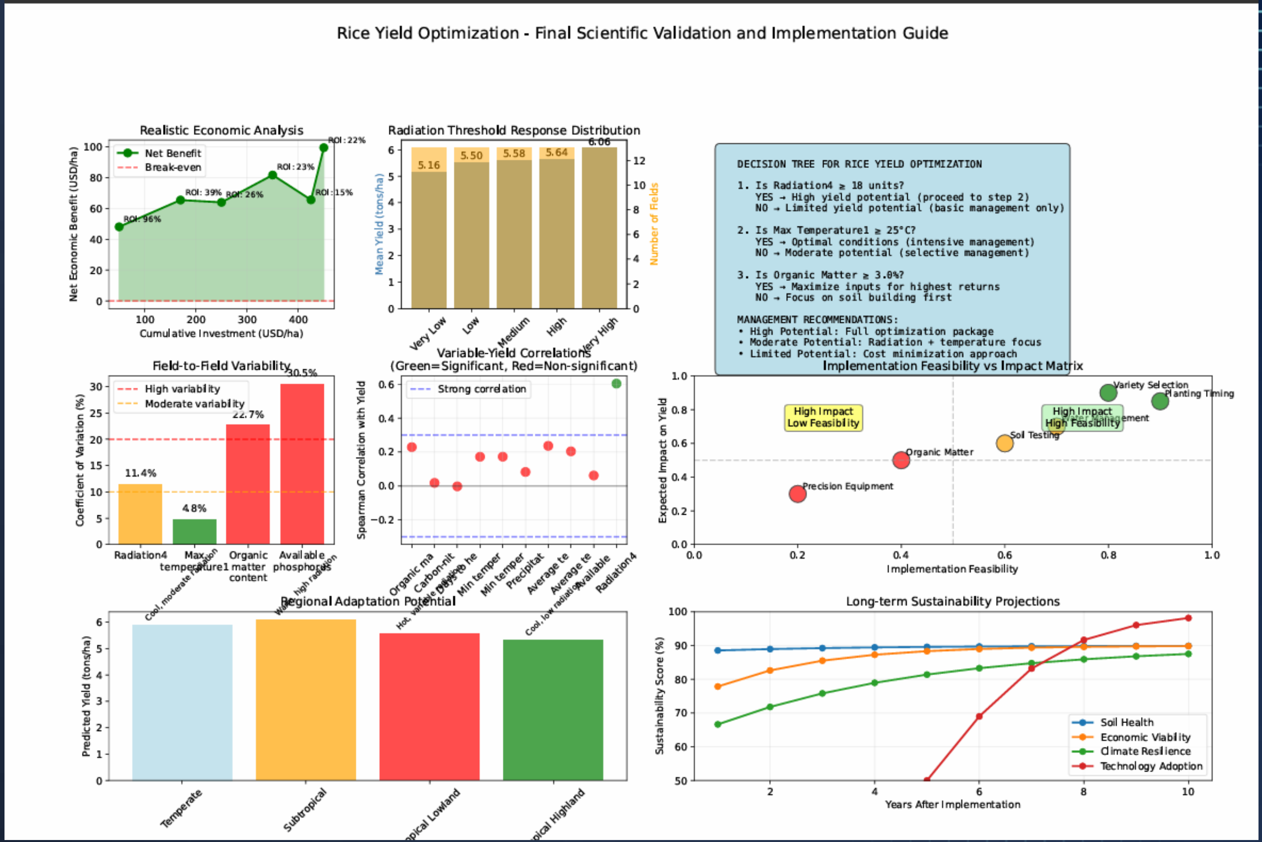
Rounds 6–7: Applicability and Risk Assessment

Round 6 – Who Can Actually Apply These Recommendations?

Objective: Assess real-world feasibility.

- Field-to-field variability
- Economic feasibility thresholds
- Regional transferability
- Resource availability constraints

Key Message: The recommendations are not universally applicable and depend on local conditions.

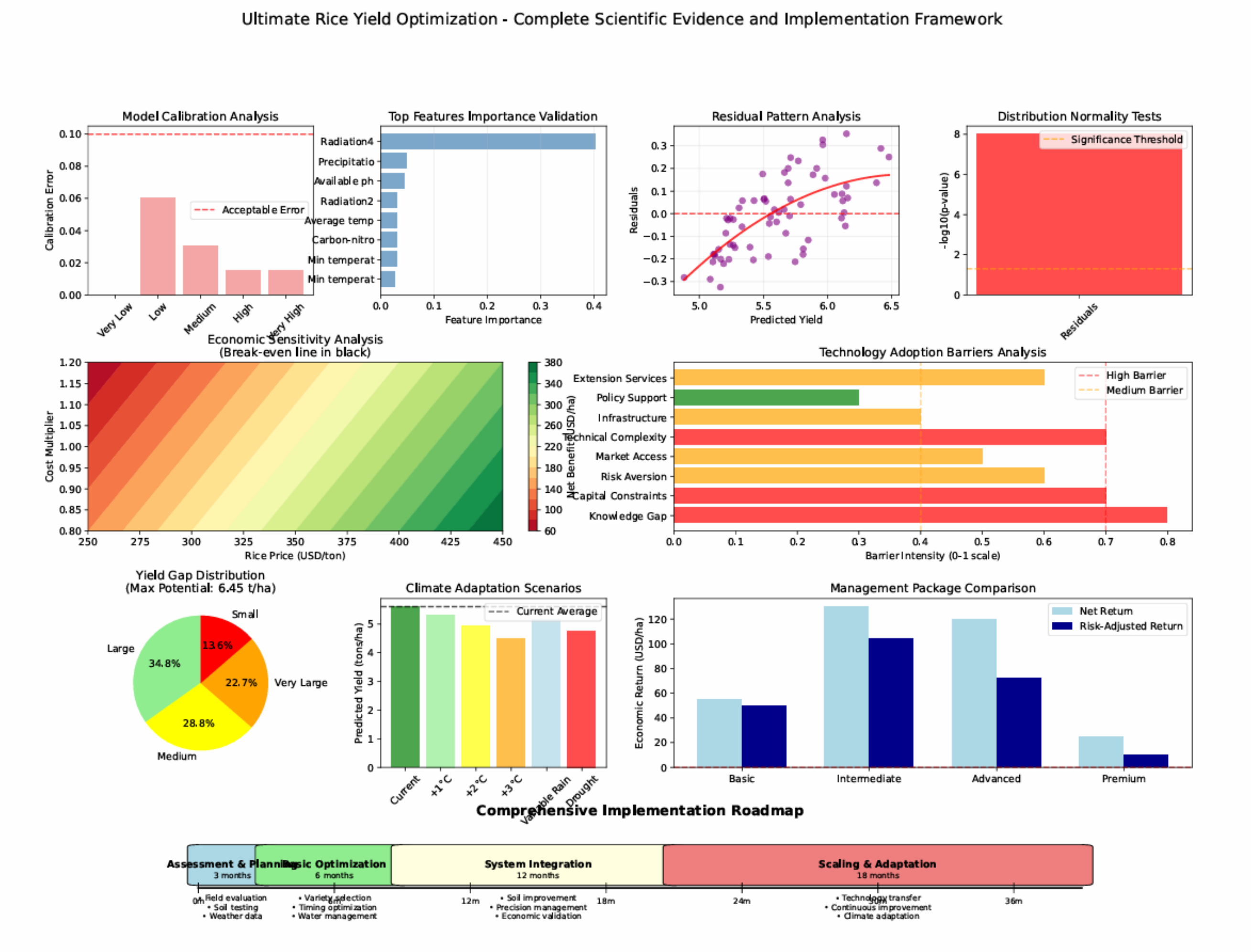


Round 7 – What Are the Risks?

Objective: Quantify uncertainty and potential failure.

- Economic sensitivity analysis
- Climate uncertainty assessment
- Technology adoption barriers
- Risk and profitability evaluation

Key Message: Success depends on specific environmental and economic conditions, and risks must be considered before implementation.



From Optimization to Realistic Decision Support

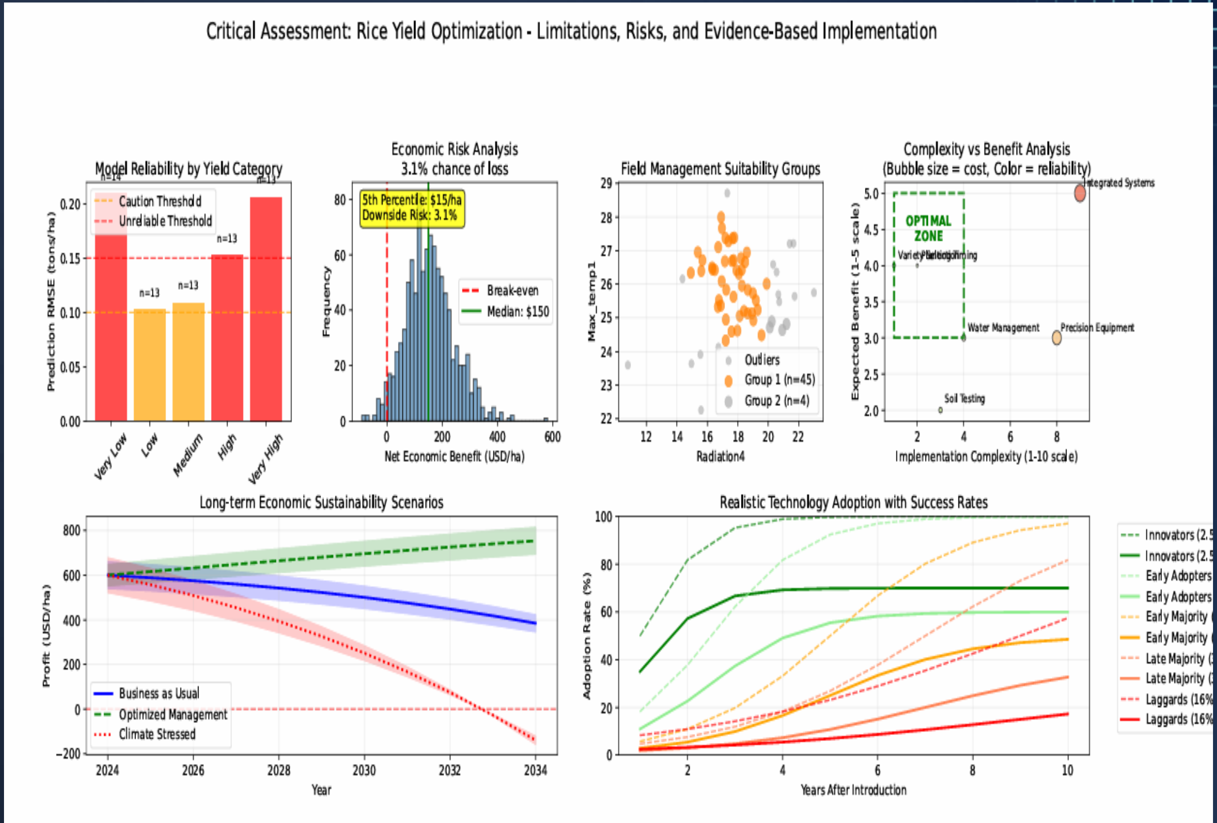
Round 8 – Failure Analysis

Question: Under what conditions does optimization fail?

- Failure modes identified
- Monte Carlo simulation performed
- Economic loss likely under many scenarios
- Not all farmers benefit equally

Key finding:

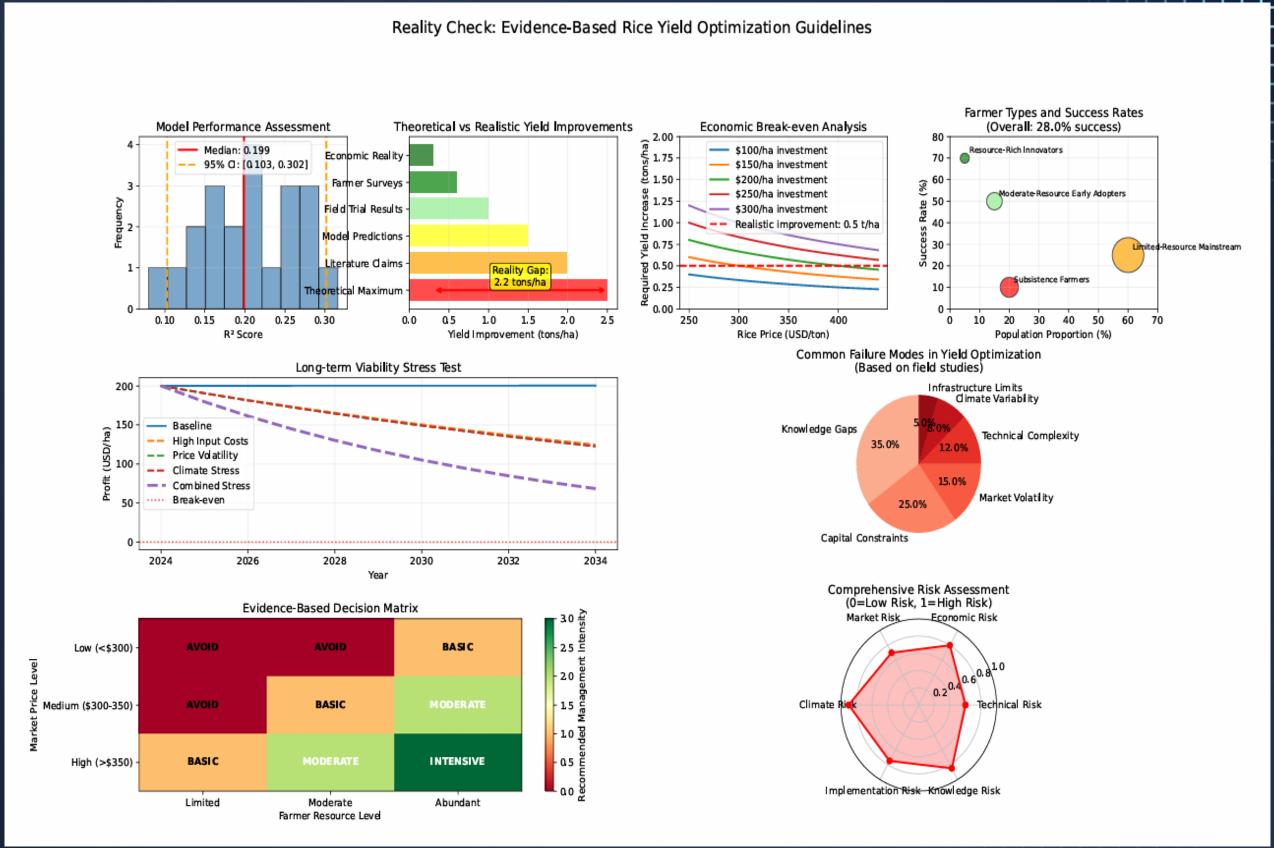
Optimization may backfire for a large proportion of farmers.



Round 9 – Reality Gap Analysis
Question: Are we overpromising?

- Compared theoretical vs practical gains
- Introduced the "Reality Gap“
- Developed a conservative decision matrix
- Focus shifted from "How to optimize" to "Should you optimize?“

Key finding: Maximum predicted yield improvements are often not achievable in real farming conditions.



From Optimization to Realistic Decision Support

Round 10 – Final Assessment
Question: Who should actually optimize?
Combined evidence from all previous rounds
Considered:

- Model uncertainty
- Economic risks
- Resource requirements
- Adoption barriers

Final conclusion
Intensive yield optimization is suitable for fewer than 2% of farmers under current conditions.